

Smart Electronics Lab v1.3

A three-kit product ladder: guided electronics lab, internet radio, and SDR workstation

Prepared for Bob Litt
Revised June 2026 — Product-Ladder Restructure

v1.3 · supersedes v1.2

0. What changed in v1.3

v1.3 restructures the product from a three-stage roadmap into a three-kit product ladder, per a new brief. The biggest change: Kit 1 no longer assumes the Elecrow has a usable audio path. Internet radio becomes Kit 1's capstone, played through a second, dedicated ESP32 audio module — and SDR/waterfall work moves entirely out of Kit 1 into Kit 2 and Kit 3. This is a restructuring, not a reversal: nearly every v1.2 technical decision (3.3V default, INA219 topology, lesson schema, fault-drill design, safety scope) carries forward unchanged inside the new kit boundaries.

Change	What happened
Roadmap shape	Three stages → three kits (Kit 1 Part 1, Kit 1 Part 2, Kit 2, Kit 3). Kit boundaries are purchase/assembly boundaries, not just curriculum boundaries.
Elecrow audio	No longer assumed. The v1.2 §4.1 "audio decision branch" is now resolved: Kit 1's internet radio always uses a second ESP32 audio module, regardless of what the Elecrow turns out to support. Confirming Elecrow I2S becomes a low-stakes curiosity, not a Kit 1 dependency.
Internet radio	Promoted from a Stage 2b appliance feature to the explicit emotional capstone of Kit 1 — the first time the learner hears sound from something they assembled.
Wi-Fi / JSON-API teaching	v1.1/v1.2's "Stage 2a" (Wi-Fi association, fetching a JSON API) had no separate home in the new ladder. It is folded into Kit 1 Part 2's lesson sequence (L12–L13), since connecting to Wi-Fi and fetching a station list is exactly what the radio capstone already requires — no separate stage needed for it.
SDR / waterfall	Moved out of Kit 1 entirely. Kit 2 is an add-on (RTL-SDR + Pi/mini PC + user's own monitor); Kit 3 is Kit 2 plus an included 15" display.
Audio subsystem choice	Researched and decided: ESP32-S3 dev board + MAX98357A I2S breakout + small speaker, on the breadboard, programmed with the Arduino framework. See §6.
Kit 2/3 compute & receiver	Researched: RTL-SDR Blog V4 recommended as the receiver; SDR++ recommended as the first-success waterfall stack. Pi 5 vs. mini PC stays an open call — 2026 DRAM-driven pricing brought them close enough that it's a real trade-off, not a default. See §13.
Kit 3 display	Researched: the official Raspberry Pi Monitor (15.6", HDMI in, USB-powered) is the default recommendation if Kit 2 compute is a Pi; a budget USB-C/HDMI portable monitor (e.g. Arzopa-class) is the fallback if compute is a mini PC. Touch not required for v1. See §14.
Command protocol	Generalized beyond knob/touch/button events to also carry audio commands (play_station, set_volume) and SDR commands (set_frequency), plus status_request. See §6 and §13.

BOM	Restructured by kit and timing instead of by stage: Kit 1–Now, Kit 1–Later (Radio), Kit 2, Kit 3. See §15.
MVP statement	Made explicit and prominent (§2): the MVP is one guided LED/resistor lesson with correct INA219 feedback — not SDR, not a 15" screen, not a proprietary breadboard.
Product tiers NEW	Kit 1 Part 1/Part 2 (build-sequence names) now also carry customer-facing tier names: Kit 1 Standard (Part 1 only) and Kit 1 Deluxe (Part 1 + Part 2). The second ESP32 audio module is reframed as a deliberate product-tier and curriculum feature, not only a workaround for the Elecrow's uncertain audio path. See §3, §6.
Future hardware path NEW	Added §19, "Future hardware convergence (v2)": a custom dial/display module that would later integrate the Elecrow's and the second ESP32 audio module's functions into one device. Explicitly deferred until after the MVP is proven; does not delay or change the v1.3 plan.
BOM completeness NEW	Restored a small "Kit 1 — Buy Later, General" table (capacitor kit, electrolytic capacitors, buzzer module) so L11's buzzer reference has a matching BOM line. These were "buy soon" items in v1.2 that fell out when trimming §15 to the brief's Buy-Now list, with no replacement row anywhere — an oversight, not an intentional cut.

No item in the new brief contradicted a v1.2 decision badly enough to flag as a blocking problem; the audio-branch resolution, the Stage-2a fold-in, and the BOM-completeness fix above are the three places this revision made a judgment call rather than just transcribing the brief, and all three are called out so they can be challenged if wrong.

1. Decision log **UPDATED**

Accepted

- Elecrow remains tutor and control head across all three kits.
- Elecrow is **not** assumed to have a usable audio output. Kit 1's internet-radio capstone always uses a second, dedicated ESP32 audio module.
- Kit 1 culminates in an internet-radio "sound moment" — this is the capstone, not an optional side feature.
- The audio module stays in the ESP32/Arduino world: ESP32-S3 dev board + MAX98357A breakout on the breadboard (§6), not the integrated ESP32-A1S/ESP-ADF path.

Open

- Exact second-ESP32 audio board SKU/vendor listing (direction set in §6; not yet sourced to a specific listing).
- Exact amp/speaker pairing and whether a small enclosure/speaker box is needed.
- Elecrow-to-audio-module physical link: UART recommended as the simplest default; I2C and Wi-Fi (e.g. ESP-NOW) remain documented alternatives.
- Kit 2 compute: Raspberry Pi 5 vs. Pi 4 vs. x86 mini PC. 2026 DRAM pricing has pushed Pi 5 and mini PCs close to cost parity, so this is a genuine trade-off now, not a default (§13).

- SDR/waterfall capability moves entirely out of Kit 1, into Kit 2 (add-on, user's monitor) and Kit 3 (Kit 2 + included 15" display).
- Raspberry Pi / mini PC compute is reserved for Kit 2/3 only — never introduced just for internet radio.
- RTL-SDR Blog V4 is the default Kit 2/3 receiver recommendation (native HF, better filtering than V3, current community default, ~\$35–40).
- SDR++ is the default first-success waterfall software for Kit 2/3 (modern, cross-platform, lighter than GQRX on a Pi).
- USB-serial JSON-per-line (carried from v1.1/v1.2) generalizes to audio and SDR commands, not just knob/touch events (§6, §13).
- First product path remains standard breadboard + smart power/measurement + display tutor — never a proprietary smart breadboard.
- 3.3V default rail, INA219 topology, extended lesson schema, structural power-off-before-rewiring, and the controlled fault-screen drill all carry forward from v1.2 unchanged.
- Kit 1 ships in two customer-facing tiers: **Kit 1 Standard** (Part 1 only) and **Kit 1 Deluxe** (Part 1 + Part 2, adding the second ESP32 audio module, I2S DAC/amp, speaker, and the internet-radio lesson pack). The tier split is a deliberate product and curriculum decision, not only a workaround for the Elecrow's unconfirmed audio path.
- A future v2 custom dial/display module is documented as a direction (§19), converging the Elecrow's and the second ESP32 audio module's functions into one device — explicitly deferred until after the MVP is proven, and explicitly not a blocker or delay for the v1.3 plan.

- Whether Kit 2 ships purely as a Kit-1-dependent add-on, or also as a partially standalone advanced kit for someone who skipped Kit 1.
- Kit 3 touch vs. non-touch 15" display — default assumption is non-touch, since the Elecrow already provides physical control (§14).
- Elecrow expansion connector type and exact pinout (carried from v1.2, still unconfirmed).
- Elecrow I2S/audio feasibility — now a low-stakes curiosity rather than a Kit 1 blocker, but still worth checking once for future flexibility.

2. The MVP, stated plainly NEW

The first MVP is **not** SDR. It is **not** a 15-inch screen. It is **not** a proprietary smart breadboard.

The first MVP is **one guided LED/resistor lesson** where the Elecrow tutor reads measured current from the INA219 and responds correctly — a pass message on expected current, a fail message on an open circuit or an out-of-range reading.

Everything else in this document is what gets built *after* that one lesson works end-to-end.

3. Three-kit product ladder

The distinctive idea is not "an electronics kit." It is a guided breadboard system where the device **teaches, observes, and reacts**. The Elecrow rotary display is the tutor throughout: it shows the lesson, accepts knob/touch input, reads smart measurement data, and guides the learner step by step — across all three kits, not just the first.

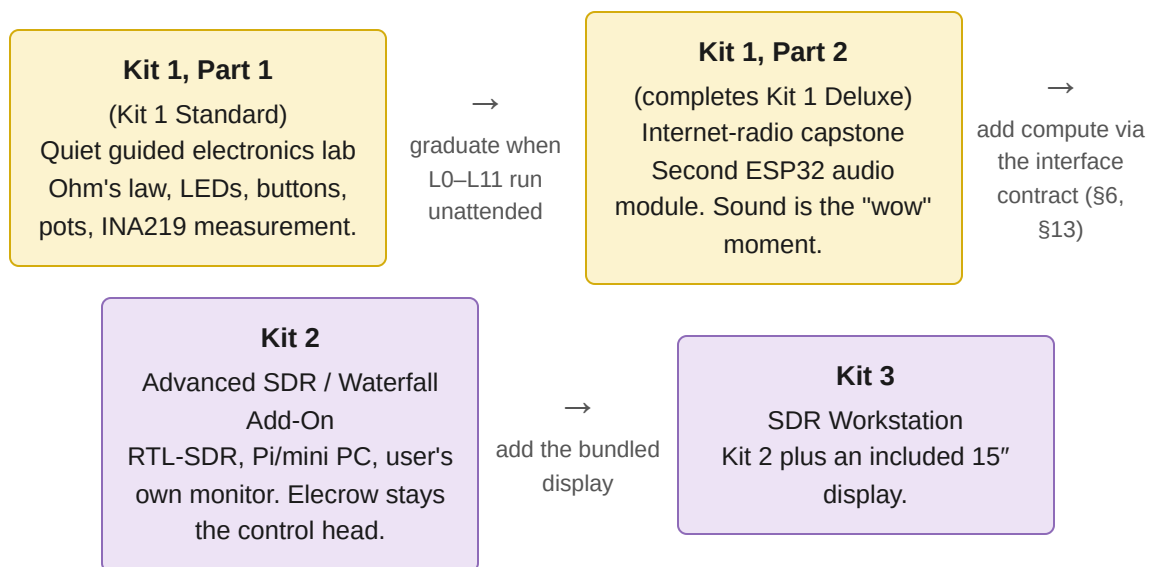


Figure 1. The three-kit product ladder. Kit 1 is one purchase with two parts; Kit 2 and Kit 3 are separate, optional purchases for users who want to go further.

Product tiers vs. build sequence: "Part 1" / "Part 2" above describe build sequence; **Kit 1 Standard** / **Kit 1 Deluxe** are the customer-facing names for the same boundary. Kit 1 Standard is

Part 1 only. Kit 1 Deluxe is Part 1 plus Part 2 — the second ESP32 audio module, I2S DAC/amp, speaker, and the internet-radio lesson pack. The split is a deliberate product and curriculum feature, not only a workaround for the Elecrow's unconfirmed audio path (§6).

Kit	Purpose	New compute?	Display	Exit criteria
Kit 1, Part 1 (Kit 1 Standard)	Quiet guided electronics lab	No — Elecrow only	Elecrow 2.1"	A beginner completes L0–L11 using only the on-screen tutor, with at least one correct INA219 current/voltage validation.
Kit 1, Part 2 (completes Kit 1 Deluxe)	Internet-radio capstone	Second ESP32 (audio only)	Elecrow 2.1"	The learner selects a saved station on the Elecrow and the second ESP32 audio module plays internet-radio audio through a speaker.
Kit 2	Advanced SDR / waterfall add-on	Raspberry Pi or mini PC	User-supplied monitor	The Elecrow knob changes an SDR parameter (tuned frequency, waterfall zoom) on the Pi/mini PC, visible on the user's monitor.
Kit 3	SDR workstation, all-in-one	Same as Kit 2	Included 15" display	Same as Kit 2, but the waterfall renders on the included 15" display.

Design rule (unchanged): every early purchase remains useful later. The breadboard teaches and prototypes. It does not become the final enclosure. The Elecrow purchased for Kit 1 is the same unit used as the control head in Kit 2 and Kit 3.

4. Open risks

None of these block ordering the Kit 1 parts list. They block committing scope for the kit they're listed under.

Risk	Why it matters	Default assumption	Resolved by
Elecrow connector/pinout unconfirmed	Determines INA219 wiring and whether an adapter cable is needed	Qwiic/STEMMA-compatible, unverified	Photograph + identify on first-boot (§17)
Elecrow I2S/audio feasibility	Was a Kit 1 blocker in v1.2; now just a future-flexibility question	Not available; not needed since Kit 1 audio moved off-Elecrow	Optional check, no deadline

Exact audio module SKU	Direction (ESP32-S3 + MAX98357A) is set; specific listing is not	Any ESP32-S3 dev board + MAX98357A breakout pair	Sourcing pass before Kit 1 Part 2 build
Elecrow ↔ audio-module link	UART, I2C, and Wi-Fi are all technically possible	UART (simplest hardware, no addressing)	First Kit 1 Part 2 build attempt
Kit 2 compute choice	Pi 5 and mini PCs are near cost parity in 2026; tradeoffs differ (ARM ecosystem vs. x86 driver coverage)	Raspberry Pi 5 (ecosystem/documentation fit); mini PC if Pi price/stock is bad	\$13 research; final call at Kit 2 purchase time
Kit 2 standalone vs. add-on	Affects whether Kit 2 needs its own onboarding/UI path	Add-on only, assumes Kit 1's Elecrow	Revisit if a no-Elecrow Kit 2 SKU is requested
Kit 3 touch display	Touch adds cost and complexity	Non-touch; Elecrow is the physical control	Revisit only if user testing shows a real gap

5. Kit 1, Part 1: hardware design

3.3V logic warning: the Elecrow / ESP32-S3 GPIO and I2C pins are 3.3V logic. Do not connect 5V directly to any GPIO, I2C, UART, touch, display, or expansion pin unless the Elecrow documentation explicitly states that pin is 5V tolerant. The breadboard may use 5V for isolated LED/resistor experiments that never touch an Elecrow pin — but any circuit connected to Elecrow data pins defaults to 3.3V logic.

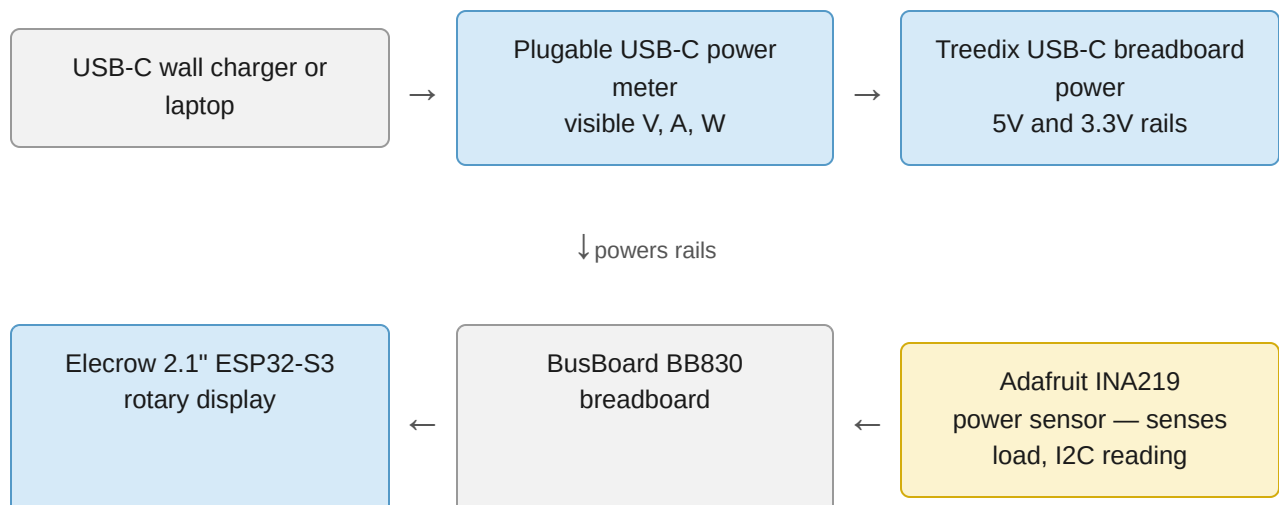




Figure 2. Kit 1, Part 1 hardware architecture (unchanged from v1.1/v1.2).

Version 0 is human-observed with the USB-C meter. Version 1 is software-observed with the INA219. Version 2 becomes a custom Smart Power Module.

Mounting answer

The Elecrow display does not need to sit in the breadboard. It sits next to the BB830 on the desk or mat, then connects to the breadboard through its GPIO/I2C connector or pigtail. The breadboard is for temporary circuits. The display is the tutor and controller.

Smart board answer

Do not start by designing a proprietary smart breadboard. Start by making the power path smart. If the Elecrow can read voltage and current from an INA219 sensor over I2C, the lesson engine can compare expected and measured behavior.

INA219 measurement topology

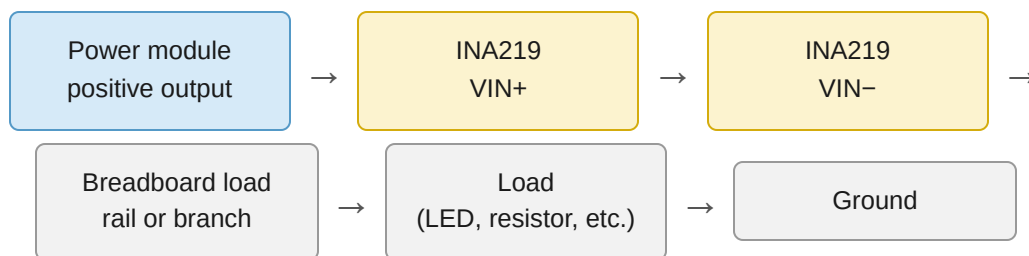


Figure 3. Correct INA219 placement — in series with the measured load, not in parallel with it. (Unchanged from v1.2.)

- The INA219 must be in the measured load's current path. If the load can draw current through any path that bypasses the INA219, the software cannot detect that current.
- The Elecrow, INA219, and breadboard circuit must share a common ground.
- The INA219's I2C logic side should be powered at the Elecrow-safe logic voltage — normally 3.3V — unless the Elecrow's I2C pins are confirmed 5V tolerant.

6. Kit 1, Part 2: internet-radio capstone NEW

Kit 1's "wow" moment is the first time the learner hears sound from something they assembled. This section assumes the Elecrow does **not** expose a clean, confirmed audio path, and routes the radio capstone through a second, dedicated ESP32 audio module instead — so the capstone never depends on an unconfirmed Elecrow capability.

Why a second module, even if Elecrow audio turns out fine: the second ESP32 audio module is not only a workaround for the Elecrow's uncertain audio path. Even if the Elecrow turned out to support clean audio output, Kit 1 Deluxe would likely still keep audio on a separate module: it keeps the tutor/control-head firmware free of streaming and decode workload, and it turns the radio capstone into a distinct, sellable tier (Kit 1 Deluxe over Kit 1 Standard) rather than a feature silently folded into the base unit.

Researched options for the audio subsystem

Option	Approx. cost	Beginner fit	Verdict
Generic ESP32 dev board + MAX98357A breakout	\$5–8 + \$4–6	High — huge tutorial base, Arduino-native	Workable, but less headroom than -S3 for future use
ESP32-S3 dev board + MAX98357A breakout	\$8–12 + \$4–6	High — same wiring as above, more PSRAM/headroom	Recommended
ESP32-A1S Audio Kit (Ai-Thinker, integrated codec)	\$10–15	Lower — less wiring, but expects ESP-ADF, which beginners consistently report as harder than the Arduino framework	Documented alternative, not the default
M5Stack Atom Echo / Core2 (integrated speaker+mic+amp)	\$13–30+	Lowest wiring burden, but appliance-like — little left to breadboard	Skips the pedagogy this kit is built around

Recommendation: an ESP32-S3 dev board plus a MAX98357A I2S amplifier breakout, wired on the breadboard, programmed with the Arduino framework (the ESP32 -audioI2S library by schreibfaul1 is the most widely used library for exactly this). Reasoning against the brief's evaluation criteria: it is the cheapest path that stays fully in the ESP32/Arduino ecosystem already used for the Elecrow; it has the deepest base of existing internet-radio tutorials and example code; the MAX98357A needs only three signal pins (BCLK, LRC, DIN — no MCLK), which keeps the wiring beginner-friendly and keeps the build on the breadboard, consistent with Kit 1 Part 1's pedagogy; and it avoids ESP-ADF, which both beginners and experienced makers report as harder to configure than the integrated ESP32-A1S board's wiring savings are worth. The ESP32-A1S and M5Stack paths remain valid documented alternatives for someone who wants less wiring at the cost of either

a steeper firmware learning curve (A1S/ESP-ADF) or a more appliance-like, less breadboard-centric build (M5Stack).

Architecture

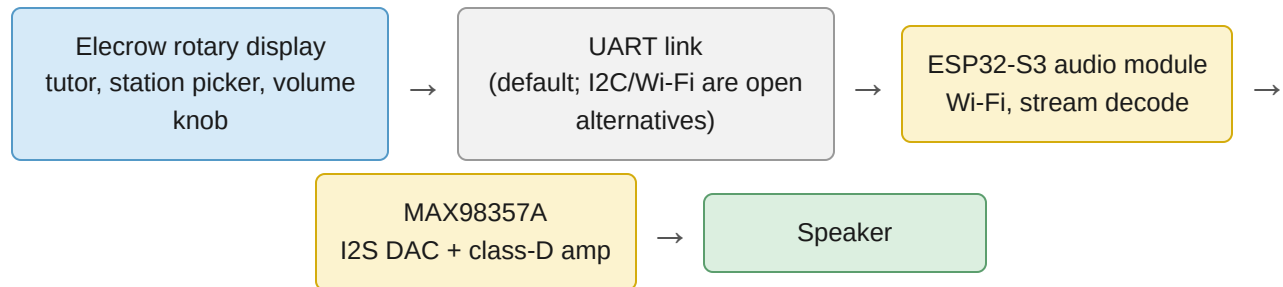


Figure 4. Kit 1, Part 2 architecture. The audio module owns Wi-Fi streaming and decoding; the Elecrow stays the tutor and the only thing the learner directly operates.

Command protocol, generalized **EXTENDED**

The v1.1/v1.2 USB-serial JSON-per-line direction carries forward, generalized to cover audio commands (and, in §13, SDR commands) alongside the original knob/touch/button events:

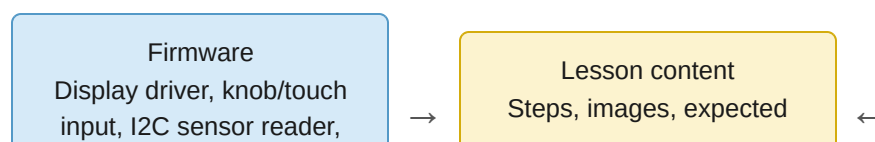
```
{"event": "knob", "delta": 1}
{"event": "button", "name": "select", "state": "pressed"}
{"event": "touch", "x": 214, "y": 93}
{"command": "play_station", "id": "station_001"}
{"command": "set_volume", "value": 42}
{"status_request": "audio"}
```

The Elecrow's firmware should already emit knob/touch/button events in this shape during Kit 1 Part 1, before the audio module exists to consume them. That is what avoids a rewrite when Part 2 is added.

What Part 2 adds to the lesson content

v1.1/v1.2's "Stage 2a" curriculum (Wi-Fi association, fetching a JSON API) did not get a separate slot in the new kit ladder — it is folded directly into the Part 2 lesson sequence (L12–L13, §11), since connecting to Wi-Fi and fetching a station list is exactly what the radio capstone already requires.

7. Software loading and lesson content



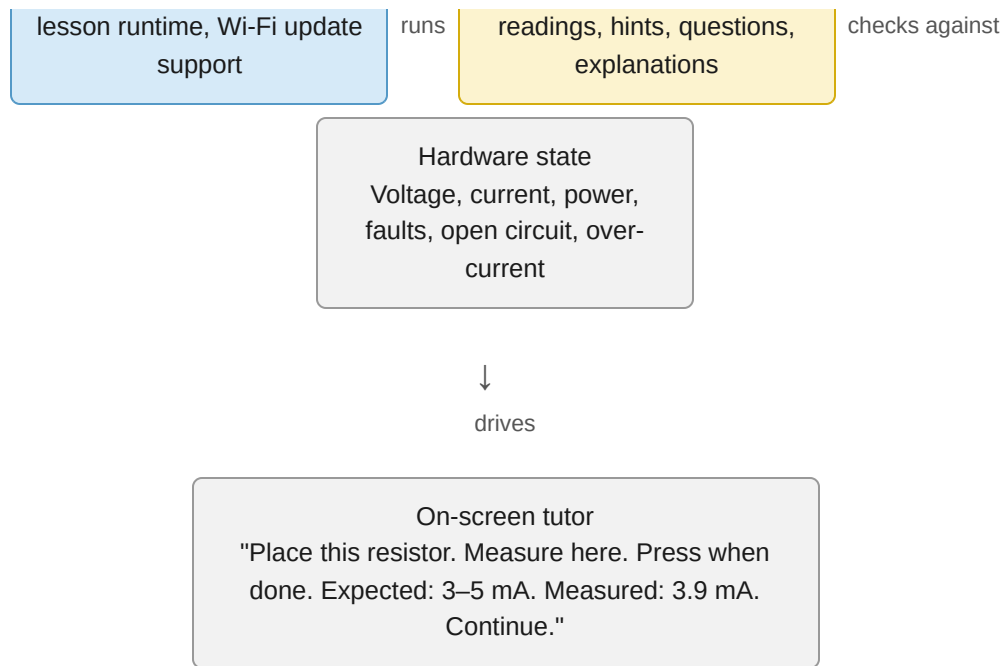


Figure 5. Software architecture for the tutor and lesson packs. (Unchanged from v1.2.)

Lesson schema

Unchanged from v1.2 — extended with safety and measurement-precision fields so assumptions are data, not prose, and with a structural power-off-before-rewiring behavior:

```
id: L2_led_with_resistor
title: "LED with resistor"
supply_rail: 3v3
steps:
  - prompt: "Place a 330 ohm resistor from row A10 to A15."
    wait_for: knob_press
    requires_power_off_before_step: true
  - prompt: "Place the LED long leg at A15, short leg to the ground rail."
    wait_for: knob_press
    requires_power_off_before_step: true
  - prompt: "Turn power on."
    measurement_source: ina219
    expected_units: mA
    expected_range: [3, 5]
    warning_threshold: 6
    hard_fault_threshold: 25
    max_safe_current_mA: 25
    gpio_safe: true
    requires_supervision: false
    on_pass: { advance: true, message: "Correct. Press to continue." }
    on_fail:
      open_circuit: "No current detected. Check the ground rail connection."
      over_range: "Current is higher than expected. Check resistor value and LED polarity"
```

hints:

- "LED current is set mostly by the resistor, not the LED."
- "Red or yellow LEDs work best at 3.3V; blue/green/white need more headroom."

next: L3_change_the_resistor

Any step with `requires_power_off_before_step: true` makes the engine render a standard interstitial — "Disconnect power before changing this wiring. Press when power is disconnected." — before the step prompt. `measurement_source` distinguishes a human-read value (`user_meter`, `usb_meter`) from a software-read one (`ina219`, `gpio_adc`). `warning_threshold / hard_fault_threshold / max_safe_current_mA` separate "this looks off" from "this is dangerous" from "this lesson's safe ceiling," and are what L10's controlled fault drill manipulates without ever exceeding the circuit's real safe operating current.

Developer mode

Connect the Elecrow to a Mac or PC with a USB-C data cable. Use PlatformIO first if possible, or Arduino IDE if following Elecrow examples. Upload firmware over USB-C. The Kit 1 Part 2 audio module is developed the same way, as a second, independent Arduino/PlatformIO project.

User mode

Later, the device should update over Wi-Fi. A user would choose a lesson pack from a menu and the device would download content. This is not Kit 1 work.

Storage plan

Do not buy a microSD card yet. The Elecrow listing specifies 16M flash, UART, I2C, and FPC expansion, but does not advertise a microSD slot. The v1 plan:

- Firmware uploaded over USB-C during development.
- Lesson content (the schema above) compiled into firmware or stored in internal flash using LittleFS or an equivalent ESP32 filesystem.
- Wi-Fi content updates deferred until the lesson engine is stable.
- External SD only if hardware inspection confirms a usable slot, or if an SPI SD module is added later.

Avoid excessive flash writes

Live measurements (voltage/current readings during an active lesson) stay in RAM. Only lesson completion, fault summaries, and calibration records get persisted to flash. High-frequency writes of live sensor data would shorten flash life for no benefit.

Lesson engine

The correct architecture is a small lesson engine, not custom firmware per lesson. The engine knows how to display a step, draw a wiring diagram, wait for a knob press, ask a question, read the current sensor, compare expected values against the schema above, and advance.

8. Smart Power Module path

V0: Visible power

Use the Pluggable USB-C meter. The learner reads voltage, current, and watts manually. Sufficient for Ohm's law lessons.

Measurement scope caveat: the Pluggable meter measures the USB-C power path, not necessarily the exact branch current of a breadboard circuit. The meter is excellent for visible V/A/W intuition; the INA219 is the correct path for software-readable, circuit-specific measurement.

V1: Software-observed power

Wire the INA219 in series with the breadboard load, connected to the Elecrow via I2C, following the topology in Figure 3 (§5). The lesson engine reads current and voltage directly.

V1 fault behavior: on over-current or out-of-range read, the lesson engine (1) shows a full-screen red fault banner, (2) displays the explicit instruction "Disconnect power now," and (3) logs the fault event (timestamp and reading, persisted). There is no automatic cutoff at V1 — the learner is the disconnect mechanism. L10 (§11) gives the learner deliberate, supervised practice with this exact banner before they can encounter it by accident.

V2: Protected smart module

Design a custom board with USB-C input, current limiting, measurement, fault reporting, and a safe disconnect circuit. This becomes the commercial product core, and is the first version where "warn" can become "warn and cut."

V3: Row-aware smart breadboard

Only after the power module works, explore sensing rows or continuity. This is much harder and should not be the first product.

9. Kit 1 desk layout

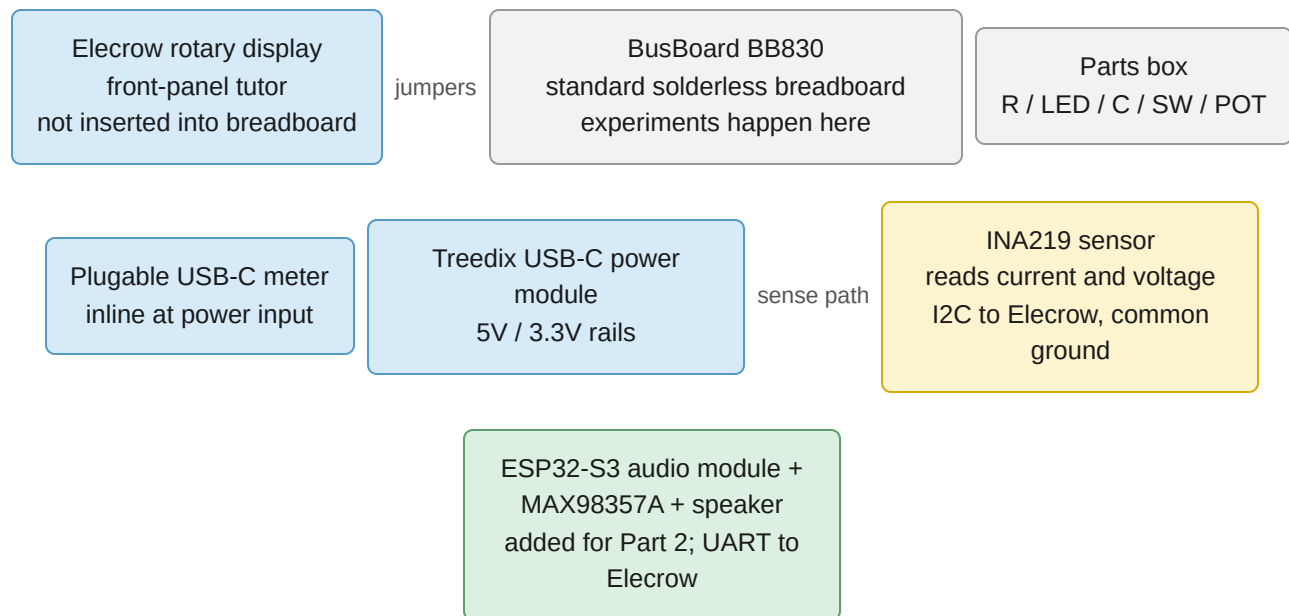


Figure 6. Recommended Kit 1 desk layout (audio module row is new in v1.3).

10. Safety scope

- Kit 1 is low-voltage DC only: 3.3V (default) and 5V (isolated experiments only) breadboard circuits.
- No AC mains. No wall outlet experiments. No exposed lithium battery charging experiments.
- **The Elecrow / ESP32-S3 GPIO and I2C pins are 3.3V logic. Never connect 5V directly to a GPIO, I2C, UART, touch, display, or expansion pin unless the documentation explicitly states 5V tolerance.** This applies equally to the Part 2 audio module's UART link.
- Assume beginner mistakes. Use current limiting, start with higher resistor values, and disconnect power before rewiring — practiced directly in L1 and enforced structurally via `requires_power_off_before_step` (§7).
- The smart module should detect over-current and shut off or warn before damage occurs. At V1, the off-the-shelf module can only warn (§8); L10 gives the learner practiced, supervised exposure to that warning before it can happen unsupervised.
- The audio module and the Elecrow must share a common ground if linked by UART or I2C; the speaker/amp side carries no GPIO-safety concerns of its own since it never touches Elecrow pins directly.

Current limits for early lessons

Lesson	Target current	Note
L2: LED with resistor	< 5 mA at 3.3V	Red/yellow LED, 330Ω–1k resistor.

L3: Resistor comparison	< 10 mA	1k / 470Ω / 330Ω at 3.3V, worst case.
L4/L5: Series & parallel	< 20 mA total	Sum across all branches.
L8: GPIO-driven LED	well under GPIO abs. max	Size the resistor conservatively (target <10 mA) regardless.
L11: Buzzer/transistor	transistor-switched only	Do not drive a buzzer directly from a GPIO pin unless its current draw is verified GPIO-safe.

11. Kit 1 lesson sequence EXTENDED, L12–L13 NEW

L0: First boot validation — before any breadboard wiring: load a vendor example over USB-C and confirm, in order: display output, capacitive touch response, knob rotation, knob press, and a successful USB-C firmware upload. Record the exact board definition, library versions, and example sketch name used.

L1: Power-off habit — the screen asks the learner to disconnect power, then walks through placing/removing a single jumper, then reconnect. No measurement yet.

L2: LED with resistor: Build a simple LED circuit at 3.3V with a red or yellow LED. The screen teaches polarity, current limiting, and expected current (target <5 mA).

L3: Change the resistor: Swap 1k, 470Ω, and 330Ω resistors at 3.3V. The screen asks the learner to predict brightness and current before measuring.

L4: Series resistance: Place two resistors in series and measure lower current.

L5: Parallel branches: Build two LED branches and observe total current increase.

L6: Button input: Wire a pushbutton and pull-down resistor. The display changes when the button is pressed.

L7: Potentiometer: Use a 10K pot as a voltage divider. Turn the pot and watch analog voltage change.

L8: Software-controlled LED: Use the Elecrow to drive an LED through a resistor. Then map knob position to brightness.

L9: Smart measurement: Insert the INA219 into the load path (Figure 3). The display compares expected and measured current.

L10: Controlled fault-screen drill — uses the same safe circuit already on the board, well within the current-limit table. The lesson temporarily lowers its own thresholds far below the circuit's actual safe current, so the existing, unmodified, safe current trips the red fault banner. No real over-current is ever created.

L11: Transistor switch: Use a transistor to drive a buzzer or higher-current LED path from a GPIO pin.

L12: Audio module bring-up NEW — wire the second ESP32-S3 + MAX98357A + speaker on the breadboard, flash a minimal Arduino sketch, and confirm a test tone plays. This is purely a hardware/firmware checkpoint — no Wi-Fi yet.

L13: Internet-radio capstone NEW — connect the audio module to Wi-Fi, fetch a small JSON station list (folding in v1.1/v1.2's old "Stage 2a" Wi-Fi/API teaching), select a saved favorite using the Elecrow knob, and send `play_station` over the UART link. Success is audio from the speaker. This is the Kit 1 finale.

12. Example on-screen lesson flow

1. Screen: Find a 330Ω resistor and a red or yellow LED. Press the knob when found.
2. Screen: Place one end at row A10 and the other at row A15.
3. Screen: Place LED long leg at A15 and short leg to the ground rail.
4. Screen: Disconnect power before connecting the 3.3V rail through the resistor path.
5. Screen: Turn power on. Expected current is about 3 to 5 mA at 3.3V with a red or yellow LED.
6. Screen, V0: Read the Plugable meter and enter or select the closest current.
7. Screen, V1: Sensor reads 3.9 mA. Correct. Press to continue.
8. Screen: Replace the 330Ω resistor with 1k. Predict whether the LED gets brighter or dimmer.

13. Kit 2: Advanced SDR / Waterfall Add-On NEW

Kit 2 is an advanced add-on for users who completed Kit 1, or who already own equivalent components. It carries no beginner electronics curriculum — it is an expansion, not a lesson pack.

Researched compute options

Option	Approx. cost (2026)	Strength	Tradeoff
Raspberry Pi 4	Lower of the two Pi options	Proven for a dedicated single-purpose receiver; huge existing RTL-SDR tutorial base	Less headroom for multiple dongles or heavier processing
Raspberry Pi 5	\$130–205 all-in depending on RAM/storage, due to 2026 DRAM price hikes	Faster CPU and USB; better fit if the station grows (more dongles, OpenWebRX, NVMe)	2026 pricing has risen sharply; no longer a clear price win over a mini PC

x86 mini PC (e.g. Beelink/GMKTec N100/N150 class)	\$170–270	Broad x86 Linux driver/software coverage — fewer "no ARM build exists" dead ends for SDR tools	Less aligned with the maker/tutorial ecosystem the rest of this project lives in
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This stays an open call (§1, §4) rather than a hard default: **Raspberry Pi 5 is the recommended starting point** for ecosystem and documentation fit with the rest of this project, with Pi 4 as a cheaper option for a single-purpose receiver and a mini PC as the fallback if Pi pricing or availability is bad at purchase time.

Researched receiver and software

RTL-SDR Blog V4 is the recommended receiver: it replaced the older R820T2 tuner with an R828D tuner plus an internal triplexer, giving native HF reception (500 kHz–1.766 GHz continuous) without an upconverter, and better filtering against FM-broadcast overload than the V3. It is the current default community answer for "what SDR should I buy," at roughly \$35–40.

SDR++ is the recommended first-success waterfall software: it is modern, cross-platform, and runs well on both Pi 4 and Pi 5 with a lighter resource footprint than GQRX. GQRX remains a documented fallback for users who prefer its simpler, more minimal Linux-native interface.

Architecture

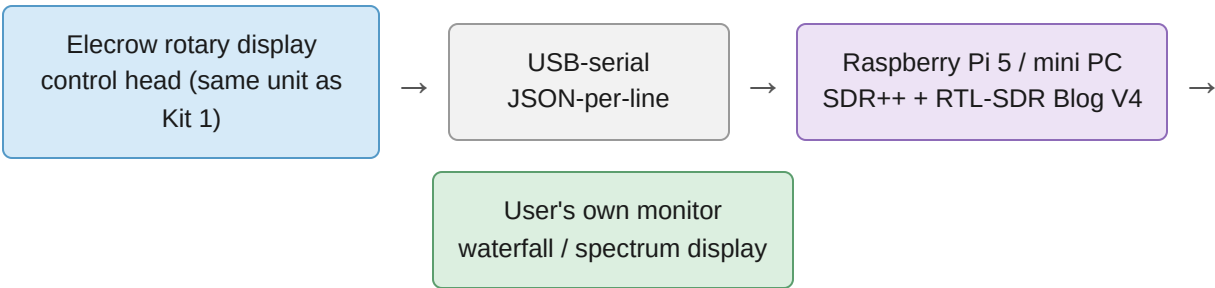


Figure 7. Kit 2 architecture — an add-on to the Kit 1 Elecrow, not a replacement for it.

What Kit 2 assumes from Kit 1, and what it adds

- **Assumes:** the Elecrow unit itself, and that its firmware already emits knob/touch/button events as JSON-per-line over USB-serial (true from Kit 1 Part 1 onward, §6).
- **Adds:** compute (Pi/mini PC), the RTL-SDR receiver, an antenna starter set, USB cables, storage, a power supply, and the SDR++ software stack. The monitor is user-supplied.

Command protocol, generalized further

```
{"command": "set_frequency", "hz": 145500000}
{"command": "set_screen", "mode": "sdr", "frequency_hz": 145500000}
{"status_request": "sdr"}
```

14. Kit 3: SDR Workstation with included 15" display NEW

Kit 3 is functionally Kit 2 plus a display: the same SDR/waterfall capability, packaged as a more complete, appliance-like product for users without a spare monitor. Higher cost, higher support burden — this is the premium tier of the ladder, not a different technical path.

Researched display options

Option	Approx. cost	Notes
Official Raspberry Pi Monitor	~\$100	15.6" Full HD IPS, standard HDMI in, USB-powered from the Pi itself. Best documentation/support fit if Kit 2 compute is a Pi.
Budget USB-C/HDMI portable monitor (Arzopa-class)	~\$70	15.6" 1080p, dual USB-C (one supports DisplayPort Alt-Mode) plus mini-HDMI. Works with either a Pi or an x86 mini PC, so it's the right fallback if compute is a mini PC.

Recommendation: the official Raspberry Pi Monitor if Kit 2's compute choice lands on a Pi; an Arzopa-class budget portable monitor if it lands on a mini PC, since the official Pi monitor is purpose-built around Pi power/mounting conventions. Touch is not recommended for v1 — the Elecrow already provides the physical control surface, and adding touch to the 15" display would duplicate that without a clear benefit.

Architecture

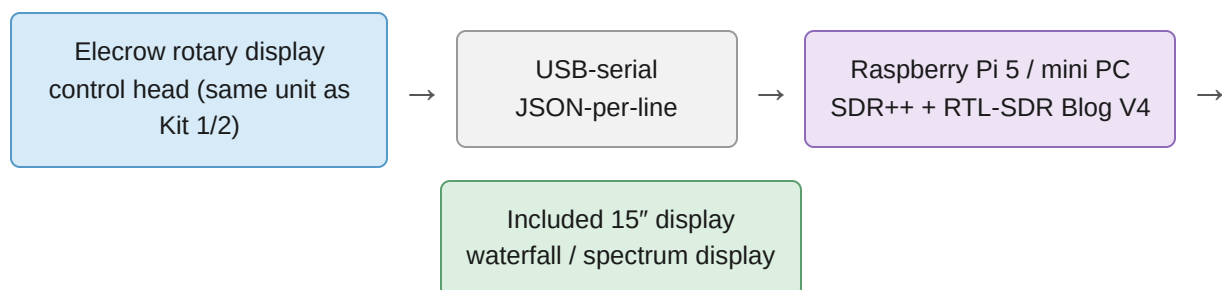


Figure 8. Kit 3 architecture — identical to Kit 2 except the display is bundled, not user-supplied.

15. Bill of materials, by kit and timing RESTRUCTURED

Durable specs live in the Interface/Spec column; ASINs and links are sourcing references only (§18), since listings rot. Items without a confirmed ASIN are described by spec; search by name/part number.

Kit 1 Standard — Buy Now

#	Item	Interface / Spec	Rationale
1	ELECROW ESP32 Rotary Display, 2.1" 480x480 IPS round (already bought)	ESP32-S3, Wi-Fi/BLE, UART, I2C, FPC expansion, 16M flash, 8M PSRAM, 5V/1A input, 3.3V logic on GPIO/I2C	Front-panel tutor and control head for all three kits.
2	BusBoard BB830 solderless breadboard	830 tie points, 4 power rails, 36V/2A rating, 21–26 AWG	Premium standard breadboard, correct size for lessons.
3	Treedix USB-C breadboard power module	USB-C in, 3.3V/5V regulated rail out	3.3V rail is the default for Elecrow-adjacent lessons.
4	Plugable USB-C Power Meter Tester	Inline pass-through, OLED V/A/W readout	Visible Ohm's law (V0). Measures the USB-C input path, not branch current.
5	Adafruit INA219 current sensor, STEMMA QT	High-side DC, 26V target, $\pm 3.2A$, I2C	Smart feedback module (V1). Must sit in series with the load.
6	Qwiic/STEMMA QT cable kit	JST-SH 4-pin to breadboard pins	Confirm Elecrow's own connector before assuming Qwiic-compatible (§4).
7	Silicone flexible Dupont jumper wires	M-M, M-F, F-F	Far less stiff than standard kit wires.
8	Aniann 1% metal film resistor assortment	1280 pcs, broad E-series coverage	Needed for every lesson.
9	BOJACK 5mm LED assortment	5 colors	Default lessons use red/yellow at 3.3V.
10	BOJACK tactile pushbutton assortment	10 values	Input lessons.
11	MCIGICM 10K breadboard trim pots	10 pcs, 10K Ω	Voltage dividers, analog input.
12	PLUSIVO BJT transistor assortment	NPN and PNP	L11 transistor switching.

13	Umirokin clear organizer boxes	2 pack, adjustable dividers	Keeps parts from becoming a loose bag.
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Kit 1 — Buy Later, General NEW

Not radio-specific. Carried forward from v1.2's "buy soon" rows so a complete Kit 1 box still includes them, and so L11's buzzer has a matching BOM line.

#	Item	Interface / Spec	Rationale
14	EMGTMS ceramic capacitor kit	24 values, 480 pcs, 10pF–10μF	RC timing, signal smoothing. Not needed for the first lessons.
15	Hilitchi electrolytic capacitor assortment	300 pcs	Power smoothing for later lessons.
16	Passive/active buzzer module pack	Arduino/ESP32 compatible	L11 (§11) drives this through a transistor, not directly from a GPIO pin, unless the module's current draw is verified GPIO-safe.

Kit 1 Deluxe — Buy Later, for the Radio Capstone

#	Item	Interface / Spec	Rationale
17	ESP32-S3 dev board (second board, audio-dedicated)	ESP32-S3, USB-C, Arduino/PlatformIO compatible	Runs the radio firmware independently of the Elecrow.
18	MAX98357A I2S amplifier breakout	I2S in (BCLK/LRC/DIN, no MCLK), 3W class-D out, 4–8Ω load	Recommended audio path — see §6.
19	Small 4–8Ω speaker, 3W class	Matched to MAX98357A output	The actual sound source for the capstone.
20	Additional jumper wires / small enclosure (optional)	—	UART link to Elecrow; speaker box is optional polish, not required for the capstone to work.

Kit 2 — Advanced SDR Add-On

#	Item	Interface / Spec	Rationale
21	Raspberry Pi 5 (or Pi 4, or x86 mini PC)	See §13 comparison — open call at purchase time	Heavy compute for FFT/waterfall.
22	RTL-SDR Blog V4 receiver	500 kHz–1.766 GHz, native HF, R828D tuner	Recommended receiver — see §13.

23	Antenna starter set	Telescopic/dipole, SDR-dongle compatible	Minimum viable reception out of the box.
24	USB cables, storage (microSD or SSD), power supply	Matched to compute choice (§13)	Standard Pi/mini-PC accessories.

Kit 3 — SDR Workstation (adds to Kit 2)

#	Item	Interface / Spec	Rationale
25	15" display (official Raspberry Pi Monitor or budget USB-C/HDMI alternative)	HDMI in, USB-powered; see §14 comparison	Bundled screen for an all-in-one workstation.
26	Display power cable, HDMI cable, stand/mount	Matched to display choice	Completes the bundled-display assembly.

16. Product concept

What is distinctive

Most electronics kits give instructions. This system gives feedback: it teaches, observes, and reacts. The device does not merely say what to build. It observes voltage and current and tells the learner what likely happened — and that same teach/observe/react loop is what makes Kit 1's radio capstone land as a "wow" moment rather than a generic feature checkbox: the learner spent Kit 1 Part 1 quietly wiring and being corrected, and the payoff is that the thing they built makes sound.

Why this can be a product

The first product does not require a proprietary breadboard, an SDR receiver, or a 15" screen (§2). The first sellable unit is a smart USB-C power and measurement module, plus an ESP32 display tutor, plus a standard BB830 — Kit 1, Part 1, full stop. Kit 1 Part 2, Kit 2, and Kit 3 are upsells to a working core, not prerequisites for it.

Product realism note: the likely first sellable product is not a smart breadboard, an SDR kit, or a workstation. It is a smart USB-C power and measurement module, paired with a standard BB830, paired with an ESP32 display tutor, with lessons as structured content packs (§7's schema). Kit 1 Part 2's radio capstone is the first upsell; Kit 2 and Kit 3 are further upsells for a smaller, more advanced segment of that same customer base.

Course packs

The same hardware could support course packs: Electronics Fundamentals, Internet Radio, Amateur Radio, SDR Basics, Sensors, Home Automation, and Robotics. With the lesson schema in §7, each course pack is a set of schema files plus assets, not a firmware fork. Internet Radio is now Kit 1's own course pack rather than a Stage 2 appliance feature; SDR Basics becomes Kit 2/3's course pack.

Commercial risk

The difficult parts are not the first hardware. The hard work is curriculum design, safe fault handling, reliable connecting, and making lessons resilient when beginners miswire circuits. The lesson schema and the V1 fault behavior (§7–8) are concrete artifacts against that risk, not just statements of it.

17. Next build action after parts arrive

1. **Photograph and identify the Elecrow connector set** that arrives in the box. Confirm 3.3V, GND, SDA, and SCL pins before connecting the INA219. Add an adapter cable to the BOM if no clean Qwiic/STEMMA-native connector is included (§4).
2. **Run L0's first-boot validation** (§11): load a basic Elecrow example over USB-C, confirm display, touch, knob rotation, knob press, and USB-C upload. Record the board definition, library versions, and example sketch name.
3. Assemble the BB830 rails with the Treedix power module. Verify 5V and 3.3V with the multimeter before connecting the Elecrow.
4. Build the first LED/resistor lesson (L2) manually using the USB-C meter for visible feedback, at 3.3V with a red or yellow LED.
5. Add the INA219 and make the Elecrow display current and voltage, following the topology in Figure 3. That is the first true Smart Lab milestone.
6. **Source and breadboard the ESP32-S3 + MAX98357A audio module** NEW and confirm a test tone plays (L12) before attempting the full internet-radio capstone (L13).
7. **Confirm the UART wiring** between the Elecrow and the audio module, and that both share a common ground.
8. Draft the Kit 2/3 control-message schema extensions (§13) as a one-page note once Kit 1 firmware's event handling is stable, so it does not need a rewrite later.

18. Sourcing references

These are retailer listings or canonical product pages, not the spec of record (see the Interface/Spec columns in §15). Prices, stock, and listing details change; if a link goes stale, search by ASIN, part name, or part number.

- **Elecrow 2.1" rotary display** (ASIN B0G3TZGRC4): rotary knob, capacitive touch, ESP32-S3, Wi-Fi/BLE, UART, I2C, FPC expansion, 16M flash, 8M PSRAM, 5V/1A input. Connector pinout and I2S availability not yet confirmed against the physical board.
- **BusBoard BB830** (ASIN B0040Z4QN8): 830 tie points, 4 power rails, 6.5 x 2.2 x 0.3 in, 36V/2A rating.
- **Treedix USB-C breadboard power module** (ASIN B0CN6HYCJD): USB-C charging/power, 3.3V and 5V outputs.
- **Plugable USB-C power meter** (ASIN B0B5W5NKKN): voltage/amperage measurement, bidirectional connection, data passthrough.
- **Adafruit INA219** (ASIN B09CBSLXN7): high-side current measurement, 26V target voltage, $\pm 3.2\text{A}$ range, I2C address options.
- **ESP32-S3 dev board + MAX98357A I2S breakout** (no single confirmed listing yet): commodity parts from many sellers (e.g. Espressif-licensed ESP32-S3-DevKitC-class boards; MAX98357A breakouts from Adafruit/SparkFun or generic 2-packs). Search by part name — do not assume a specific ASIN until sourced.
- **RTL-SDR Blog V4** (rtl-sdr.com store / official RTL-SDR Blog listing): R828D tuner, native HF via internal triplexer, 500 kHz–1.766 GHz, ~\$35–40.
- **Raspberry Pi 5** (raspberrypi.com): current pricing inflated by 2026 DRAM costs; compare against mini PC options (e.g. Beelink/GMKTec N100/N150 class) at purchase time.
- **Raspberry Pi Monitor, official** (raspberrypi.com/products/raspberry-pi-monitor): 15.6" Full HD IPS, HDMI in, USB-powered. Default Kit 3 display if compute is a Pi.
- **Adafruit INA260** (official page): higher-capacity future option for a custom V2 Smart Power Module, integrated shunt, I2C, 3V/5V logic compatible, up to 36V/15A.

19. Future hardware convergence (v2) NEW

Not a v1.3 dependency: nothing in this section changes or delays the near-term plan. v1.3 stays off-the-shelf and buildable today — Elecrow + breadboard for Kit 1 Standard, plus a second ESP32 + MAX98357A + speaker for Kit 1 Deluxe. This section documents a future direction, gated on proving the MVP first. It is not a design task to start now, and it has no BOM line in §15.

Kit 1 Deluxe currently splits its functions across two boards: the Elecrow (tutor, control head, knob/touch UI) and a second ESP32 audio module (Wi-Fi streaming, I2S decode, amplification). That split is the right v1 architecture — cheap, off-the-shelf, and it keeps each board's firmware simple. It is also, by construction, a preview of a single future device.

Trigger condition

Once the MVP is proven — the lesson engine, the smart measurement loop (INA219 path, §5/§8), and the internet-radio capstone (§6) all working end-to-end on off-the-shelf parts — it becomes worth designing a custom dial/display module that converges what the Elecrow and the second ESP32 audio module do today into one board.

What the custom module should include

- Rotary control (carried from the Elecrow's knob interaction model).
- Touchscreen (carried from the Elecrow's capacitive display).
- A protected expansion connector, so breadboard/sensor wiring mistakes can't easily damage the main board.
- USB-C power and data, for both charging/power and firmware updates.
- A built-in smart-measurement interface (the INA219-class current/voltage sensing path, §5/§8), instead of routing it through an external breakout.
- A clean audio path — an onboard codec, an integrated I2S amplifier, or at minimum a speaker connector — so the radio capstone no longer needs a second physical board.

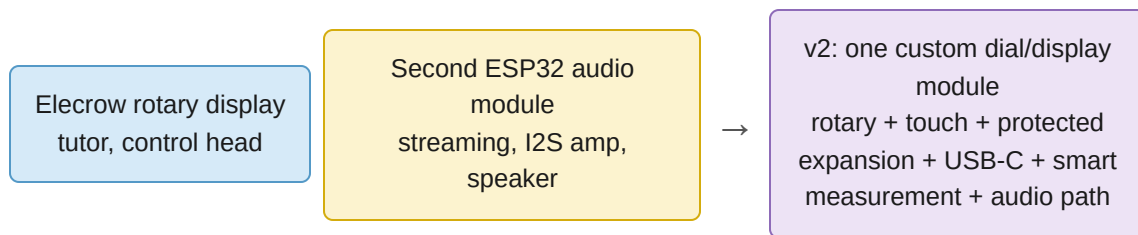


Figure 9. v2 convergence direction — two off-the-shelf boards today, one custom module later, only after the MVP is proven.

This is a hardware roadmap entry, not a current build task: no part numbers, no BOM line, and it should not be started before Kit 1 Standard and Kit 1 Deluxe both work reliably with off-the-shelf parts.